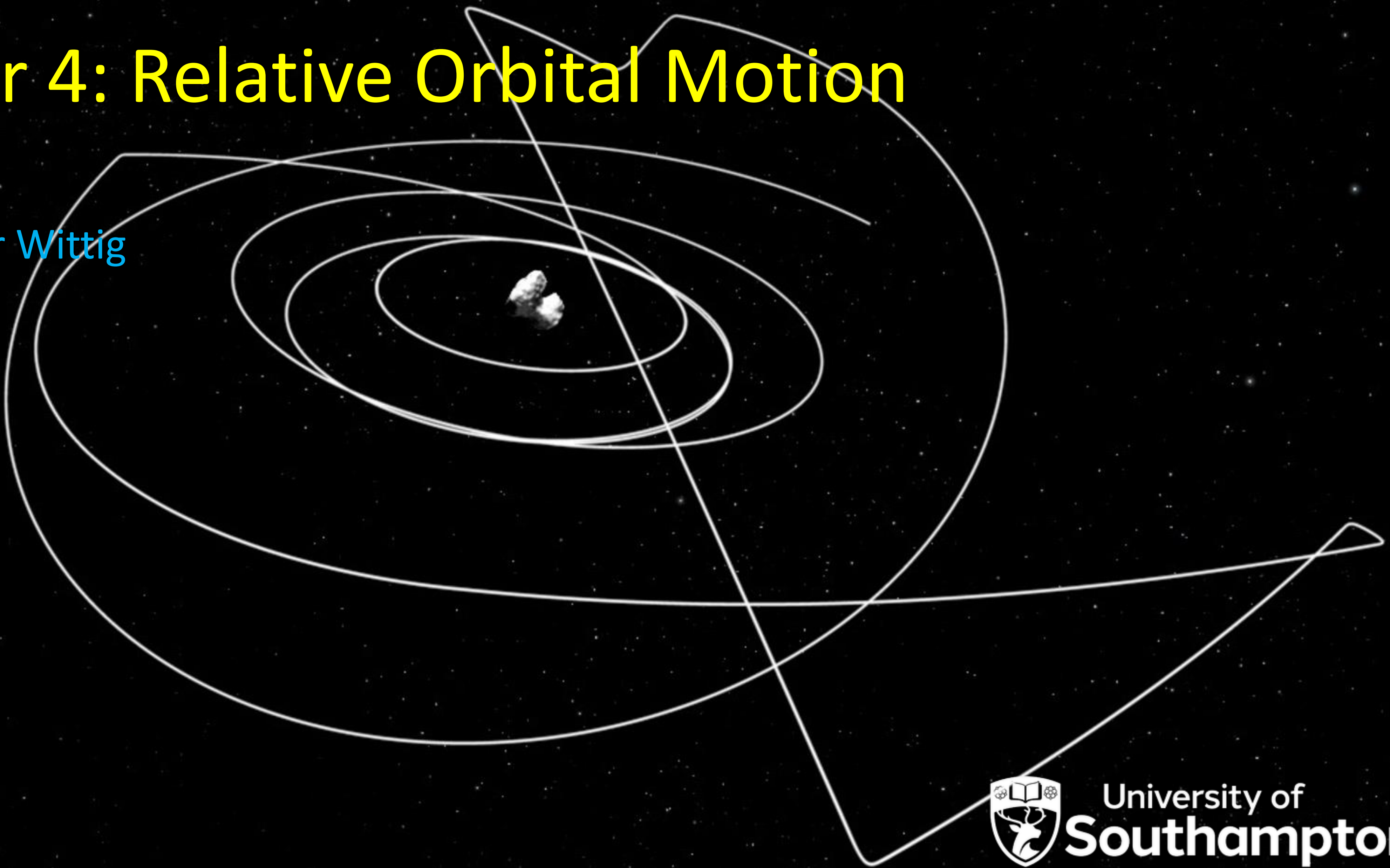
Rosetta lander deployment orbit (ESA)



Chapter 4: Relative Orbital Motion

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Questions

- Any questions on previous weeks content?

Course Content

- **Introduction**
- **Chapter 1: In 20.000 km, turn right.**
  Orbital Manoeuvres
  - Coplanar Manoeuvres
  - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing**
  Orbital Perturbations & Propagation
  - Non-Uniform Gravity Field
  - Third-Body Perturbation
  - Atmospheric Drag
  - Solar Radiation Pressure
  - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird?**
  Orbit Observation & Determination
  - Observation Techniques
  - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda**
  Relative Orbital Motion
  - Relative Orbital Motion
  - Linearization & State Transition Matrix
- **Chapter 5: To boldly go…**
  Interplanetary Trajectories & Optimization
  - Lambert's Problem
  - Patched Conics
  - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies**
  Restricted Three-Body Problem
  - N-Body Problem
  - Circular Restricted Three-Body Problem
  - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round**
  Attitude Dynamics
  - Attitude representation
  - Attitude Dynamics
- **Chapter 8: In the spin cycle**
  Attitude Motion
  - Torque-Free Motion
  - Attitude Maneuvers
- **Summary and Revision**

Learning outcomes

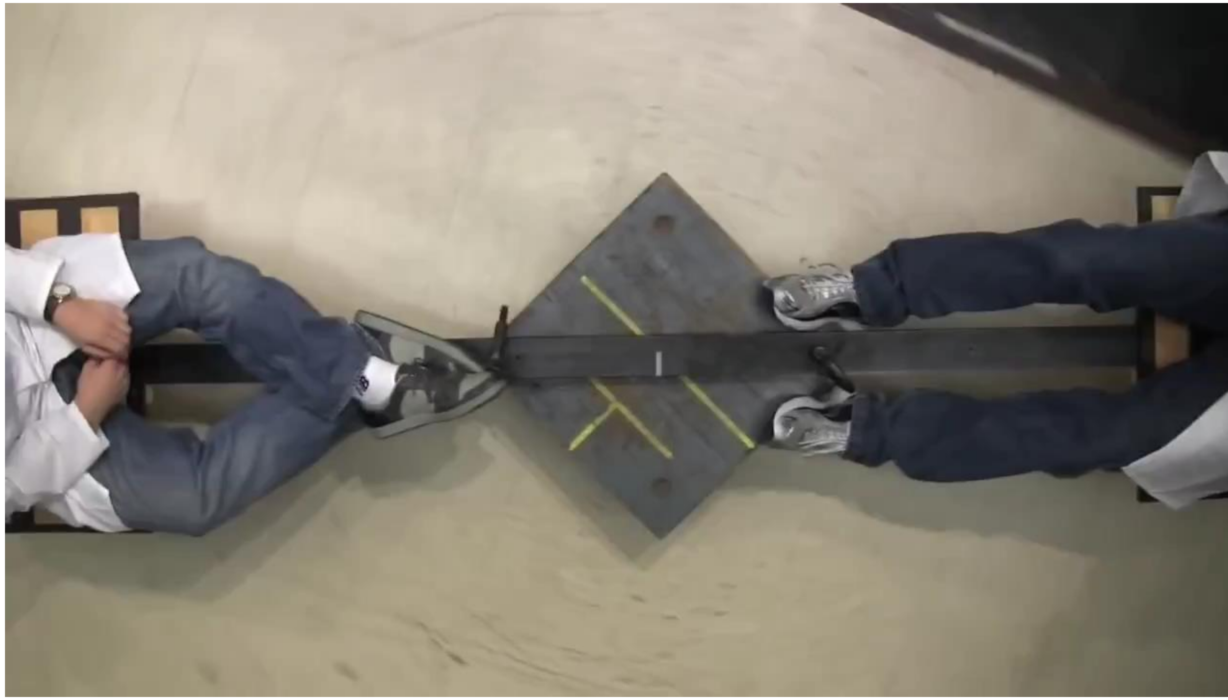
- Define the LVLH frame
- State assumptions made to arrive at Clohessy-Wiltshire eqns
- Calculate relative motion for Earth orbiting satellites using CW eqns
- Sketch and intuitively explain some typical orbits resulting from CW equations

Relative Orbital Motion

Derivation

Relative motion

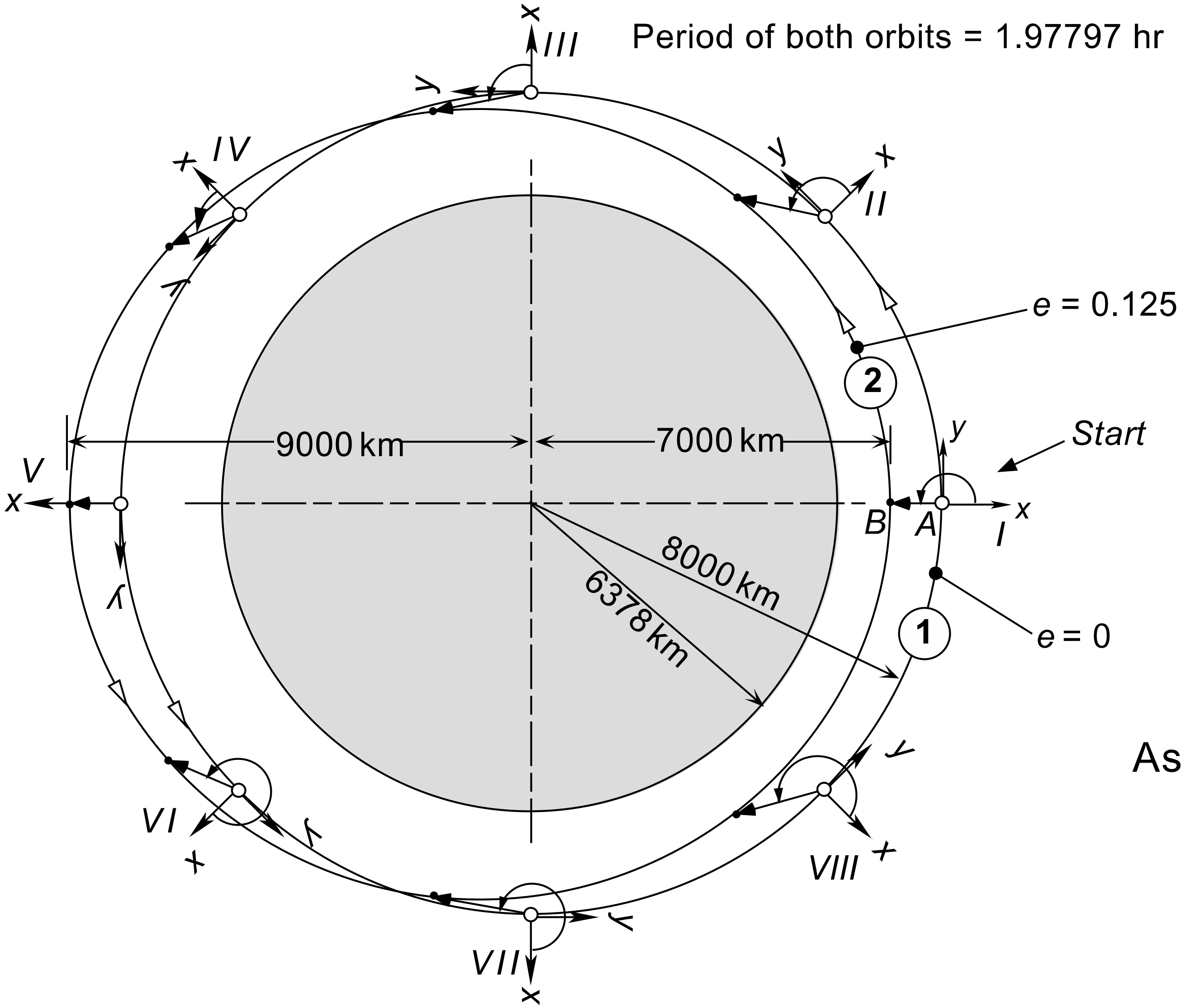
MIT Dept. of Physics



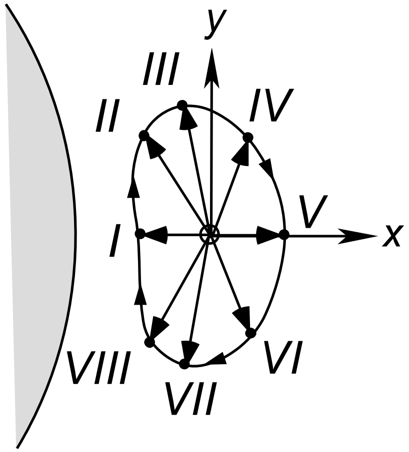
Relative motion in non-accelerating frame
simple, along straight line

Relative motion in accelerating frame (e.g. rotating)
unintuitive, curved paths

Example of relative motion in space



As viewed in the inertial frame.



As viewed from the co-moving frame in circular orbit **1**.

Curtis

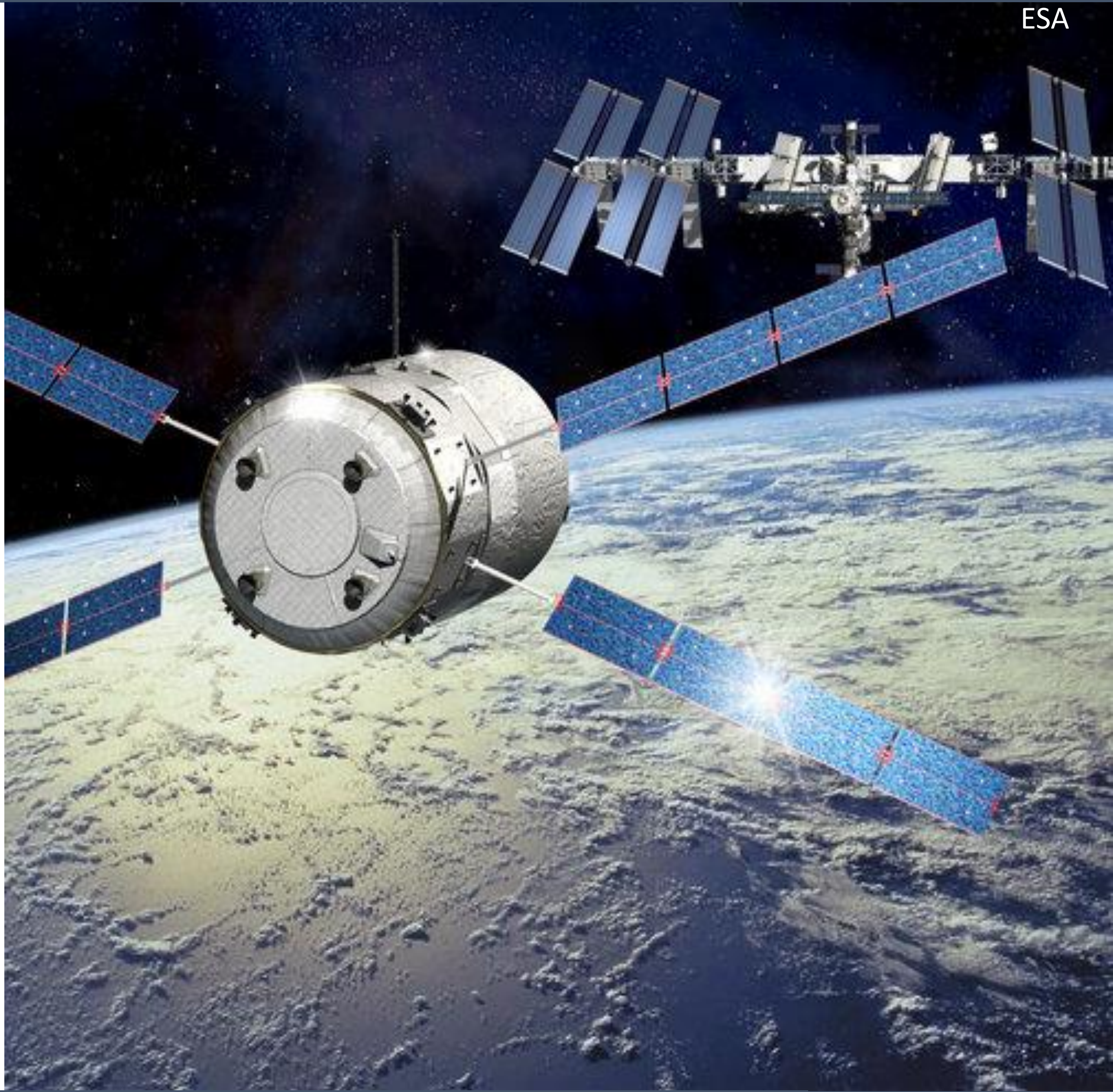
Why do we care?

- Rendezvous and docking maneuvers
  - ISS & ESA ATV
  - Space Shuttle & Hubble
  - In orbit servicing
  - Active space debris removal

- Uncertainty quantification
  - "Virtual" target = reference orbit
  - Actual spacecraft state

Space X ISS docking simulator:

https://iss-sim.spacex.com/


ESA

Margaret Hamilton

Career:

- BA in Mathematics (Earlham College, 1958)
- Programmer (MIT, 1959)
- Director Software Engineering (Draper Labs)

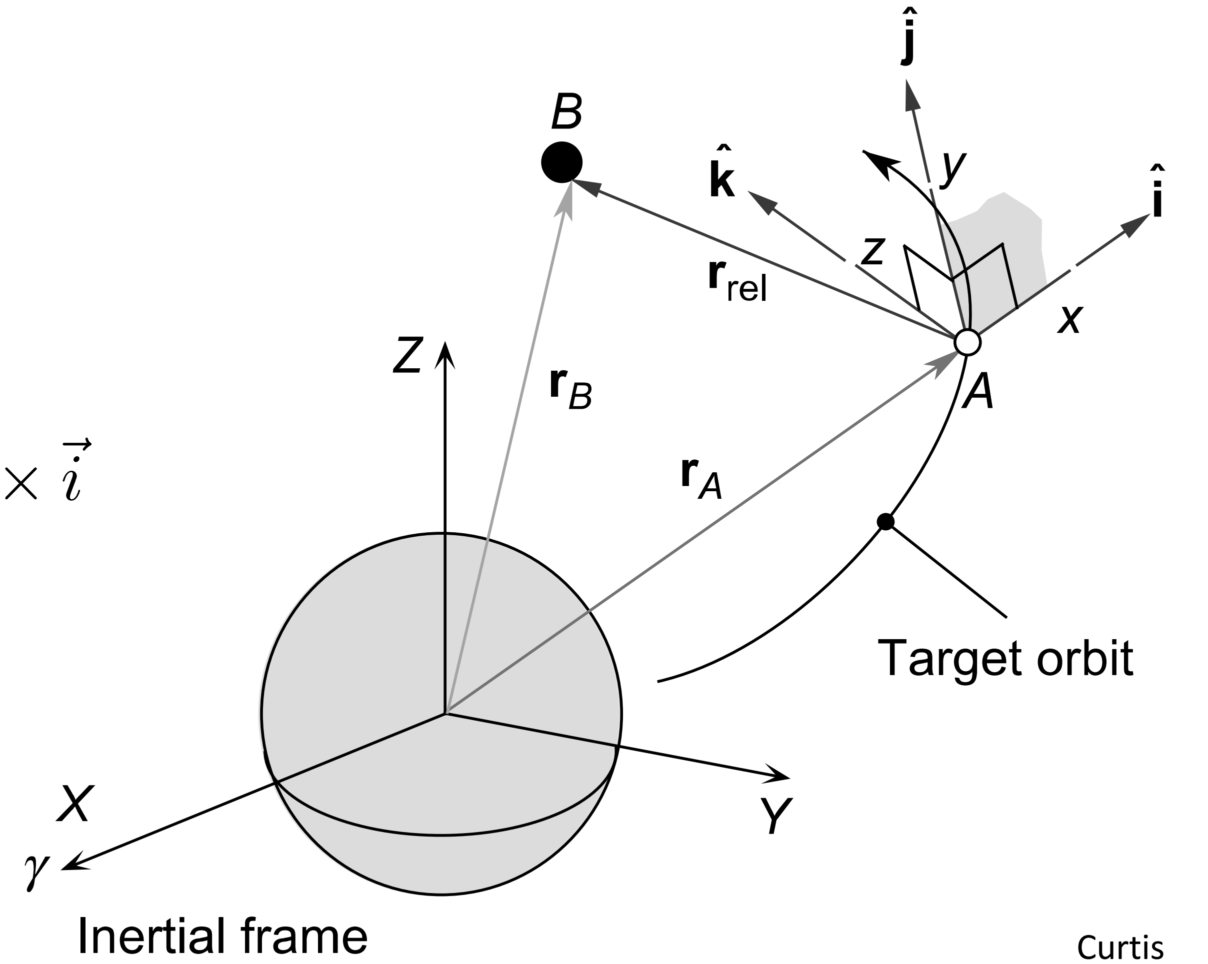
Achievements:

- Developed Apollo 11 guidance computer
- Pioneered error handling in flight software
  - ”never supposed to happen” display
  - prioritizing and scheduling
  - rigorous software testing
- Invented field of “software engineering”

MIT News

LVLH frame

- Local vertical local horizontal frame
- Frame attached to (target) SC
  - x (**i**) along radial
  - z (**k**) out of orbital plane
  - y (**j**) completes right-handed frame $\vec{j} = \vec{k} \times \vec{i}$
- Origin co-moving with target
- Rotating with target ➔ not inertial (via radial direction)



Curtis

LVHL frame

- Local vertical local horizontal frame

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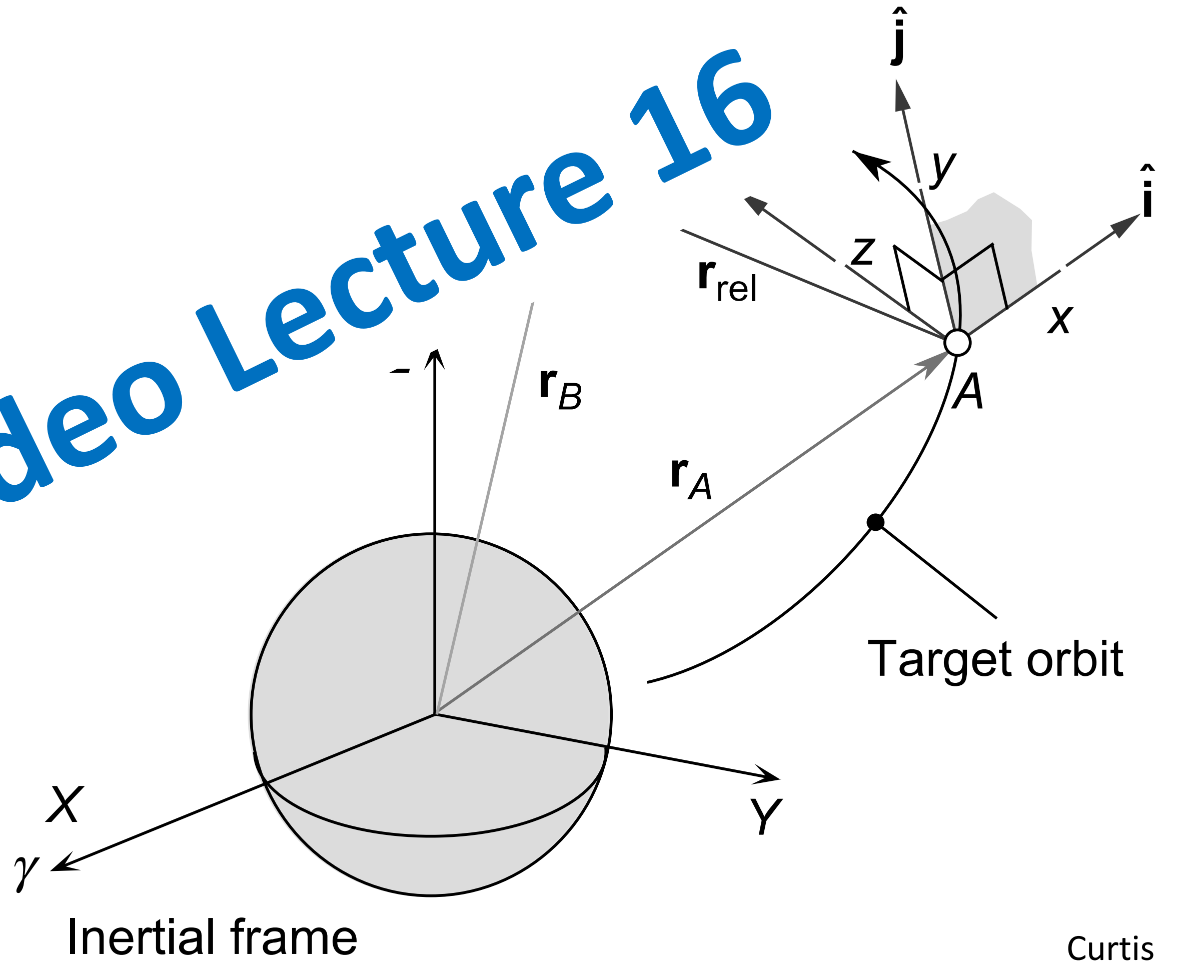
see Chapter 5 Video Lecture 16



Curtis

Relative equations of motion

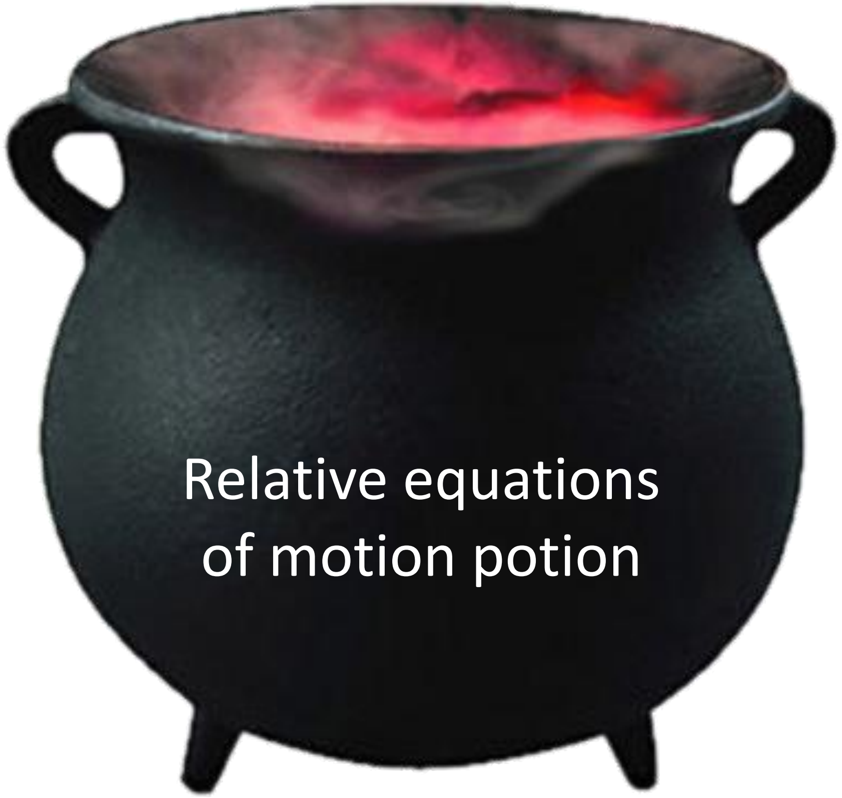
Velocity and acceleration in rotating frame

Angular velocity & acceleration of co-moving frame

Velocity and acceleration in inertial frame

Transformation from inertial to rotating frame

Linearization

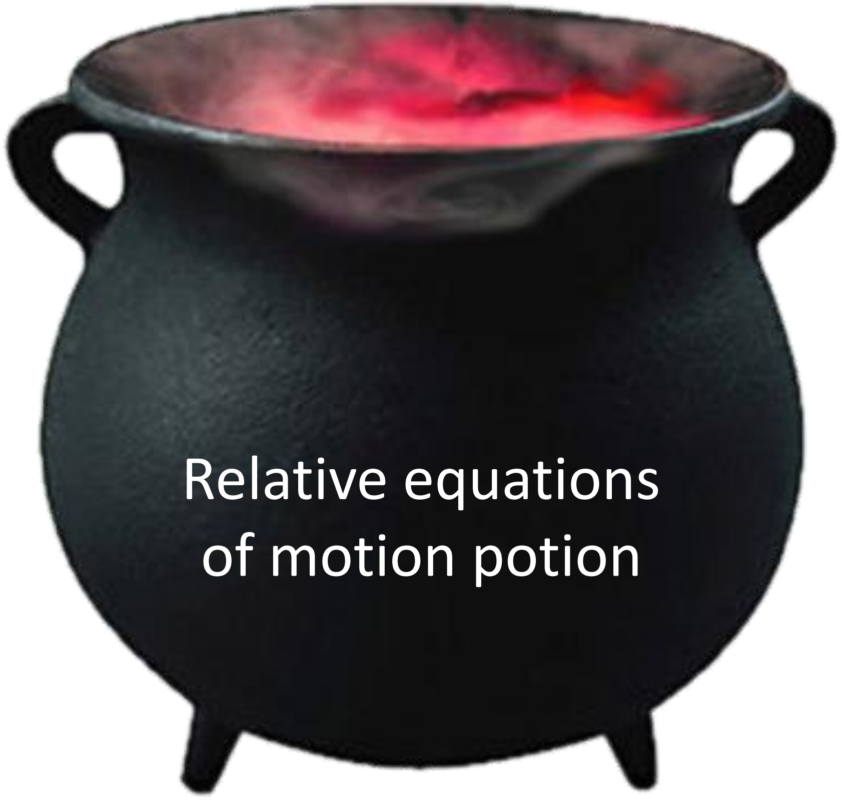


Relative equations of motion

Result: ODE for relative coordinates of chaser in LVLH frame of target.

Given:

- orbital motion of target (position and velocity as function of time)

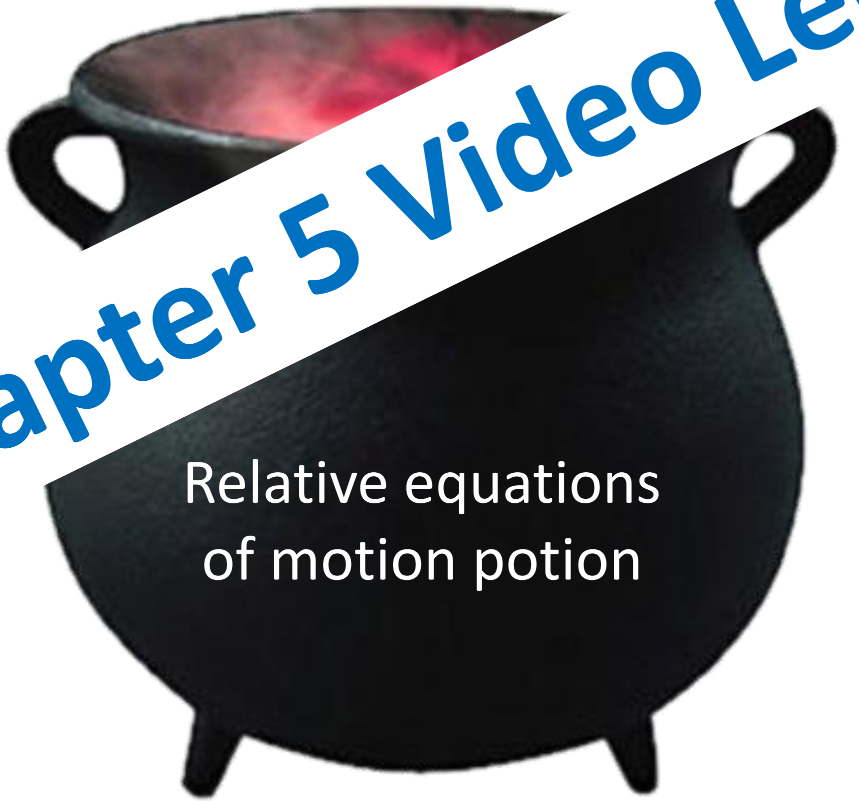


Result: ODE for relative coordinates of chaser in LVLH frame of target.

Given:

- orbital motion of target (position and velocity time)

see Chapter 5 Video Lecture 17



Final equations of relative motion

$$\delta\ddot{x} - \left(\frac{2\mu}{R^3} + \frac{h^2}{R^4}\right)\delta x + \frac{2(\mathbf{V}\cdot\mathbf{R})h}{R^4}\delta y - 2\frac{h}{R^2}\delta\dot{y} = 0$$

$$\delta\ddot{y} + \left(\frac{\mu}{R^3} - \frac{h^2}{R^4}\right)\delta y - \frac{2(\mathbf{V}\cdot\mathbf{R})h}{R^4}\delta x + 2\frac{h}{R^2}\delta\dot{x} = 0$$

$$\delta\ddot{z} + \frac{\mu}{R^3}\delta z = 0$$

$\delta x, \delta y, \delta z :$ relative chaser coordinates (LVLH frame)

$\boldsymbol{R}, \boldsymbol{V} :$ position and velocity of target (relative to primary)

$h :$ angular momentum of target orbit

Simplifying Assumptions

The following assumptions are made for CW equations:

- LVLH frame relative to **target S/C**
- Relative **distances are small** compared to target orbital radius (linearization!)
- Target is in **circular** orbit ← additional assumption

Must validate all assumptions before using CW!

Result: Analytical solution known as the Clohessy-Wiltshire equations

Analytical solution: Clohessy-Wiltshire Equations

Linear in initial conditions, so can be written in matrix form:

$$\{\delta\mathbf{r}(t)\} = [\mathbf{\Phi}_{rr}(t)]\{\delta\mathbf{r}_0\} + [\mathbf{\Phi}_{rv}(t)]\{\delta\mathbf{v}_0\}$$

$$\{\delta\mathbf{v}(t)\} = [\mathbf{\Phi}_{vr}(t)]\{\delta\mathbf{r}_0\} + [\mathbf{\Phi}_{vv}(t)]\{\delta\mathbf{v}_0\}$$

$$[\mathbf{\Phi}_{rr}(t)] = \left[\begin{array}{cc|c} 4-3\cos nt & 0 & 0 \\ 6(\sin nt - nt) & 1 & 0 \\ \hline 0 & 0 & \cos nt \end{array}\right]$$

$$[\mathbf{\Phi}_{vr}(t)] = \left[\begin{array}{cc|c} 3n\sin nt & 0 & 0 \\ 6n(\cos nt - 1) & 0 & 0 \\ \hline 0 & 0 & -n\sin nt \end{array}\right]$$

$$[\mathbf{\Phi}_{rv}(t)] = \left[\begin{array}{cc|c} \frac{1}{n}\sin nt & \frac{2}{n}(1-\cos nt) & 0 \\ \frac{2}{n}(\cos nt - 1) & \frac{1}{n}(4\sin nt - 3nt) & 0 \\ \hline 0 & 0 & \frac{1}{n}\sin nt \end{array}\right]$$

$$[\mathbf{\Phi}_{vv}(t)] = \left[\begin{array}{cc|c} \cos nt & 2\sin nt & 0 \\ -2\sin nt & 4\cos nt - 3 & 0 \\ \hline 0 & 0 & \cos nt \end{array}\right]$$

Z motion

What is the z motion over time given only a small initial perturbation in z?

$$\{\delta \mathbf{r}(t)\} = [\boldsymbol{\Phi}_{\mathrm{rr}}(t)]\{\delta \mathbf{r}_0\} + [\boldsymbol{\Phi}_{\mathrm{rv}}(t)]\{\delta \mathbf{v}_0\}$$

$$[\boldsymbol{\Phi}_{\mathrm{rr}}(t)] = \left[\begin{array}{cc:c} 4 - 3\cos nt & 0 & 0 \\ 6(\sin nt - nt) & 1 & 0 \\ \hdashline 0 & 0 & \cos nt \end{array}\right]$$

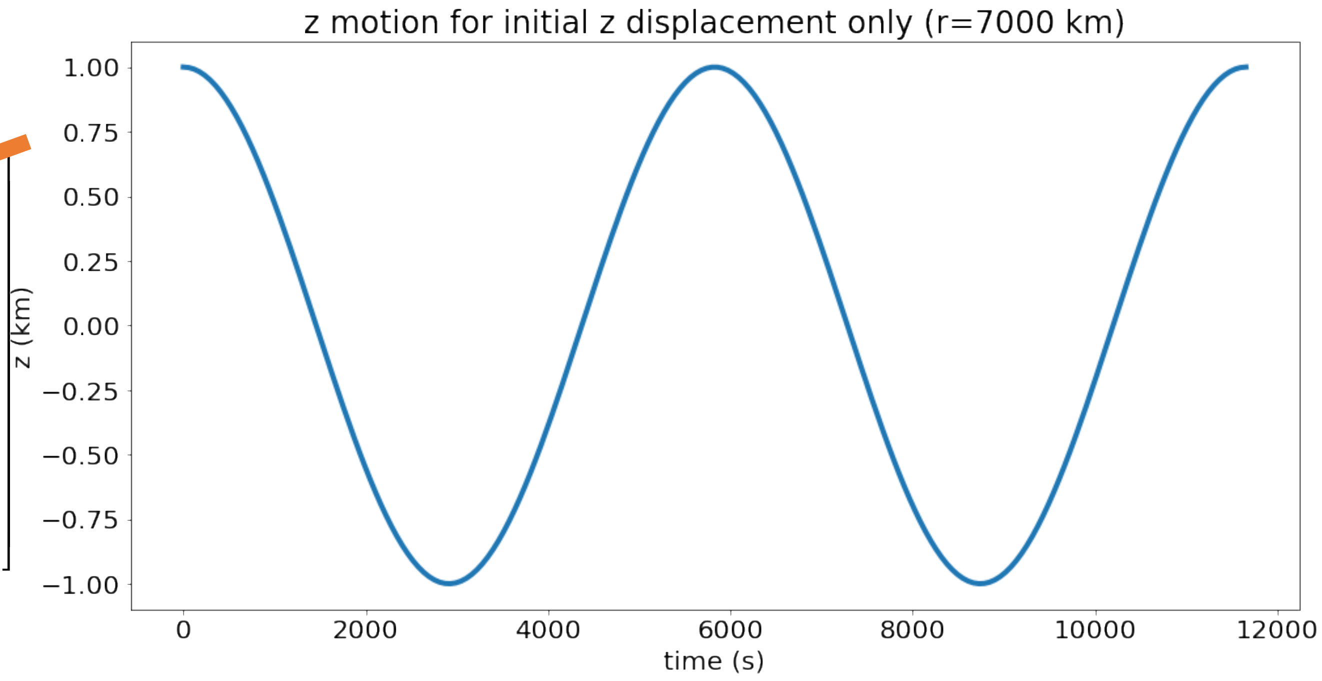
$$[\boldsymbol{\Phi}_{\mathrm{rv}}(t)] = \left[\begin{array}{cc:c} \frac{1}{n}\sin nt & \frac{2}{n}(1 - \cos nt) & 0 \\ \frac{2}{n}(\cos nt - 1) & \frac{1}{n}(4\sin nt - 3nt) & 0 \\ \hdashline 0 & 0 & \frac{1}{n}\sin nt \end{array}\right]$$

$$\delta \vec{r}_0 = \begin{pmatrix} 0 \\ 0 \\ \delta z_0 \end{pmatrix}$$

$$\delta x(t) = 0$$

$$\delta y(t) = 0$$

$$\delta z(t) = \delta z_0 \cos(nt)$$



What is the x-y motion given only a small initial perturbation in x?

$$\{\delta \mathbf{r}(t)\} = [\mathbf{\Phi}_{rr}(t)]\{\delta \mathbf{r}_0\} + [\mathbf{\Phi}_{rv}(t)]\{\delta \mathbf{v}_0\}$$

$$[\mathbf{\Phi}_{rr}(t)] = \left[\begin{array}{cc|c} 4 - 3\cos nt & 0 & 0 \\ 6(\sin nt - nt) & 1 & 0 \\ \hline 0 & 0 & \cos nt \end{array}\right]$$

$$[\mathbf{\Phi}_{rv}(t)] = \left[\begin{array}{cc|c} \frac{1}{n}\sin nt & \frac{2}{n}(1 - \cos nt) & 0 \\ \frac{2}{n}(\cos nt - 1) & \frac{1}{n}(4\sin nt - 3nt) & 0 \\ \hline 0 & 0 & \frac{1}{n}\sin nt \end{array}\right]$$

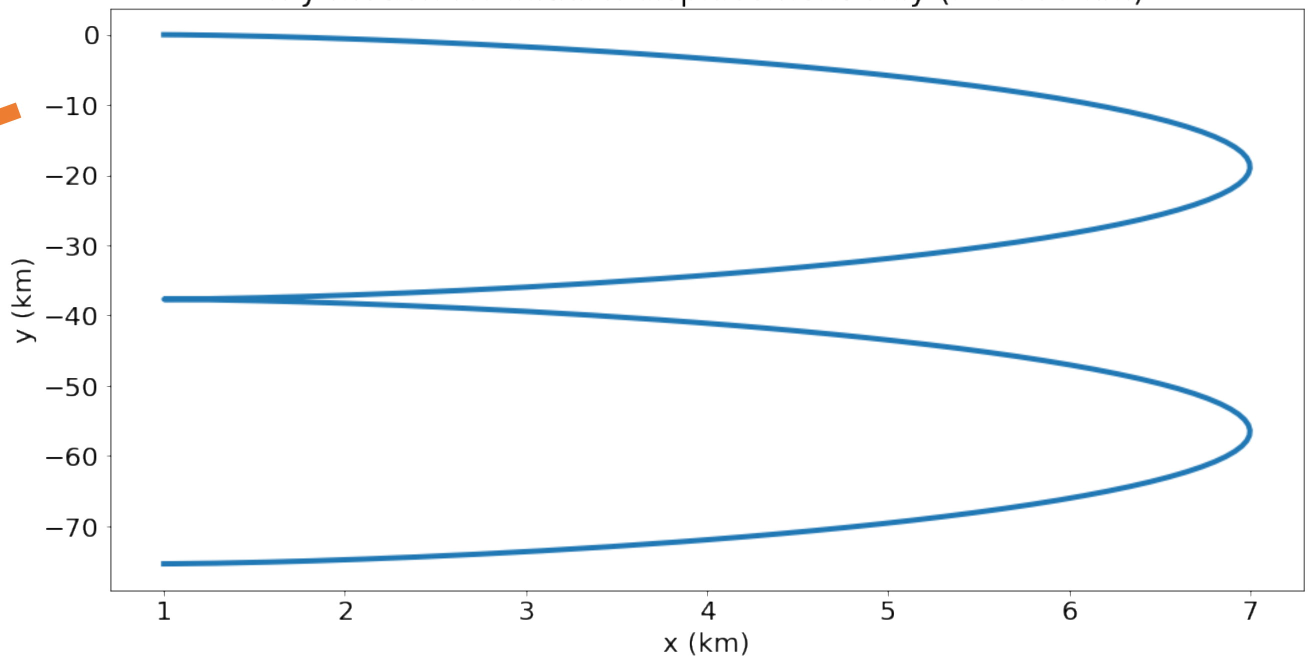
$$\delta \vec{r}_0 = \begin{pmatrix} \delta x_0 \\ 0 \\ 0 \end{pmatrix}$$

$$\delta x(t) = \delta x_0 \left(4 - 3\cos(nt)\right)$$

$$\delta y(t) = 6\delta x_0 \left(\sin(nt) - nt\right)$$

$$\delta z(t) = 0$$

x-y motion for initial x displacement only (r=7000 km)



Qualitative behavior of CW

To understand motion, consider position evolution for a few simple cases

Good skill to be able to sketch these **qualitatively**!

- Out of plane motion (z-direction)
- Displacement in x direction only
- Displacement in y direction only
- Displacement in x velocity only
- Displacement in y velocity only

Linear: effect of multiple displacements is simply additive

Qualitative behavior of CW

To understand motion, consider position evolution for a few sim… cases

Good skill to be able to sketch these **qualitatively**!

- Out of plane motion (z-direction)
- Displacement in x direction only
- Displacement in y direction on…
- Displacement in x veloc…
- Displacement in … …y

see Chapter 5 Video Lecture 18

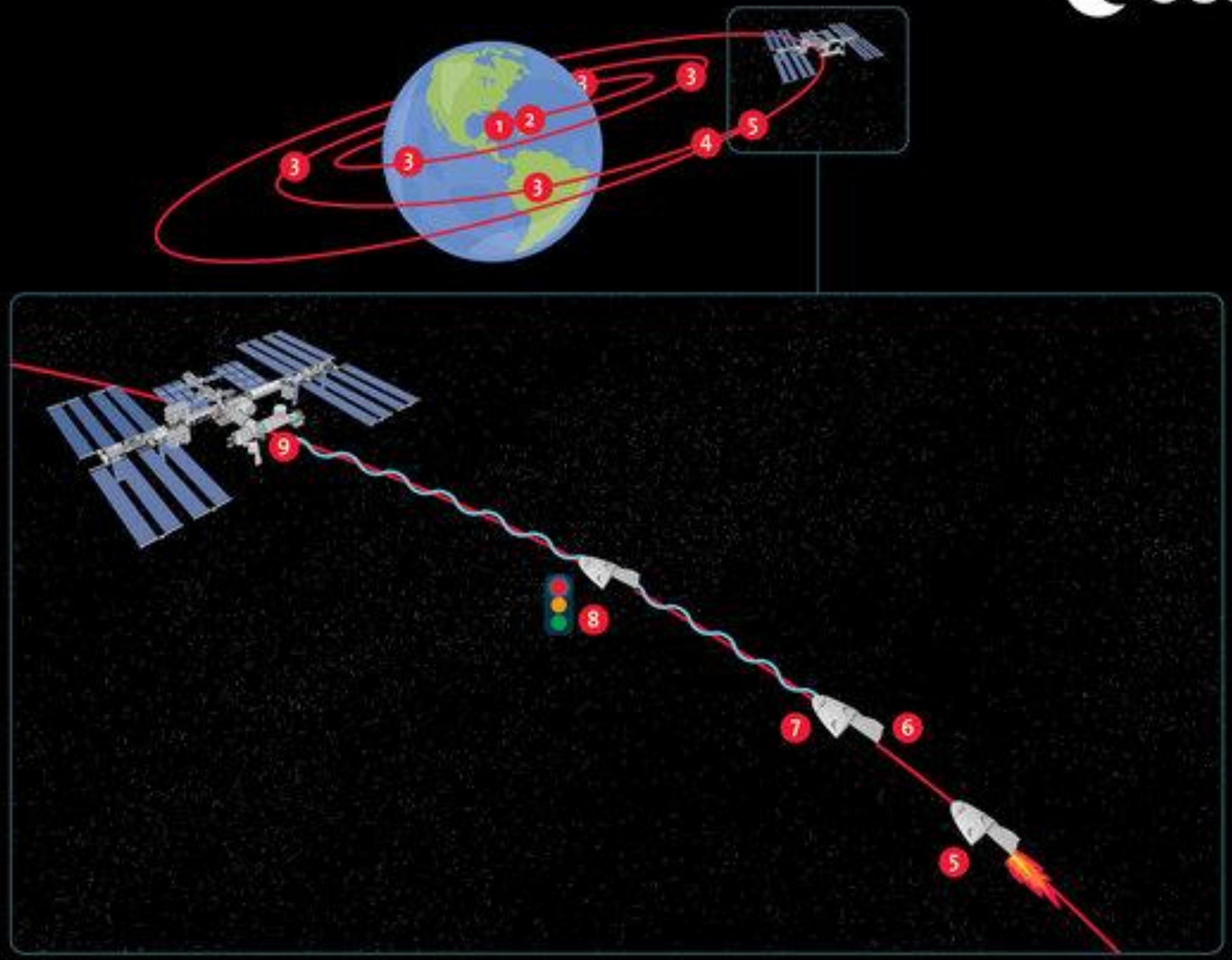
Linear: effe… …ultiple displacements is simply additive

Autonomous docking

CREW DRAGON RENDEZVOUS AND DOCKING

The Crew Dragon takes around 18 hours to reach the International Space Station while orbiting Earth.

1. Liftoff.
2. Separation.
3. Engines fire five times during the day to catch up with the International Space Station.
4. Dragon establishes communication link.
5. 7.5 km behind and below the Space Station, Dragon fires engines for final approach.
6. 200 m from the Space Station, Dragon establishes relative navigation.
7. The final approach is autonomous from 200 m out.
8. Dragon holds position at 20 m from the Space Station and awaits go from Mission Control.
9. Dragon docks with a free International Docking Adapter port on the Space Station.



Example Problem

The dragon capsule is located at some location relative to the ISS.

What initial relative velocity does it need to dock (without further maneuvers) after a given time t?

Example Problem

The dragon capsule is located at some location relative to the ISS.

What initial relative velocity does it need to dock (without further maneuvers) after a given time t?

$$\delta r(t) = \mathbf{\Phi}_{rr}(t)\delta r_0 + \mathbf{\Phi}_{rv}(t)\boxed{\delta v_0}$$

Summary

- Relative motion measured relative to target LVLH frame (co-rotating, co-moving → non-inertial)
- Linearize relative acceleration of chaser: final ODE => numerically integrated
- Simplification: target on circular orbit (CW equations) => analytical solution
- Qualitative behavior for perturbations in different directions
- Useful for rendezvous & docking, sensitivity analysis

Further reading

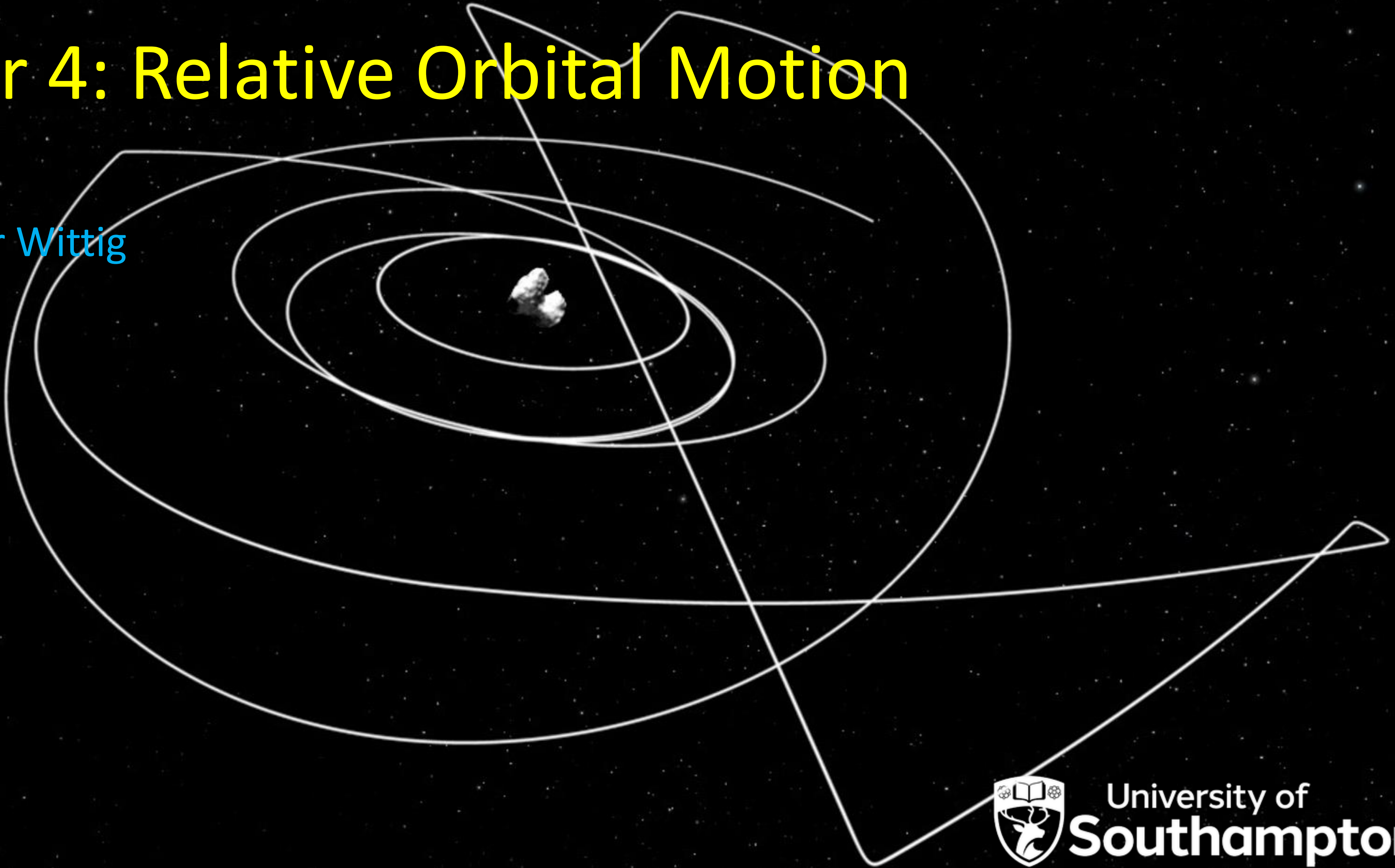
- Curtis, Chapter 7

Also:

- Vallado, Chapter 6.8

Chapter 4: Relative Orbital Motion

Dr. Alexander Wittig



Rosetta lander deployment orbit (ESA)